

Sai Krishna Are

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Summary

MS in Data Science candidate & Data Engineer with 2.5+ years of experience designing and building scalable ETL pipelines, automating data workflows, and ensuring data integrity across distributed systems. Skilled in constructing data lakes and warehouses, implementing Airflow for orchestration, and maintaining cloud-native infrastructure on AWS. Strong background in scripting, data validation, and CI/CD automation. Adept in cross-functional collaboration and mentoring junior team members. Passionate about delivering high-quality, reliable data systems to support analytics and business intelligence.

Education

University of Houston, MS in Engineering Data Science Aug 2023 – May 2025

- GPA: 3.5/4.0
- **Coursework:** Data Science, Machine Learning, Big Data Analytics, Data Mining, Database Management Tools, Health Analytics and Visualization (Using Tableau)

Experience

Software Engineer, Renault Nissan Mitsubishi Alliance – Chennai, INDIA Jan 2021 – Aug 2023

- Worked as a Software Engineer in Platform Services team, managing CI/CD deployments and automating 40+ feature releases of Teamcenter and PSI application in Production.
- Designed and implemented scalable ETL pipelines and Dataflow to automate data integration between cloud storage systems (AWS S3, JFrog Artifactory).
- Built automated data pipelines and scheduling workflows using Airflow for improved data integration.
- Deployed and maintained Linux-based virtual machines on AWS EC2 Engine for hosting log management and data monitoring services.
- Implemented CI/CD pipelines in GitLab, reducing manual tasks and increasing deployment efficiency to 80%.
- Designed and optimized SQL queries for efficient data extraction and reporting using MySQL Workbench.
- Developed PowerShell and Bash scripts to monitor application usage and automate system alerts.
- Developed Unix shell scripts to synchronize the data between multiple servers based on Solaris OS.
- mentored a team of 5 engineers in troubleshooting, automating server-side tasks, and data integration.
- Conducted rigorous data validation and built anomaly detection scripts to ensure data integrity.
- Actively participated in Agile/Scrum and Kanban environments using tools like Jira and Confluence, contributing to sprint planning, backlog grooming, and daily stand-ups..
- Maintained comprehensive documentation using Confluence for deployment workflows, automation scripts, and system configurations to support scalable development and knowledge sharing.
- Applied version control best practices using Git, ensuring collaboration and codebase integrity across multi-functional development teams.Managed incident tracking and task management using Jira and ServiceNow, ensuring timely issue resolution and service requests.
- Managed access to cloud resources securely using AWS IAM roles and policies, ensuring fine-grained permission control for team members

Data Engineering Projects

Next Gen. Image Segmentation and Object Detection for Medical Imaging. Feb 2025 - May 2025

- This project is a deep learning solution for automated brain tumor detection and segmentation in multimodal MRI images. Developed a deep learning pipeline for automated brain tumor detection using U-Net and MobileNetV2.
- Managed model training and data preprocessing using TensorFlow and OpenCV.

Job Hiring Portal - Job Nest

Feb 2024 - May 2024

- Job Nest is a dynamic web-based job hiring portal designed to streamline the recruitment process by bridging the gap between job seekers and employers. Designed and managed relational data models for user and job profiles using SQL. Implemented backend logic for ingesting, storing, and retrieving structured job seeker and employer data.
- Tools Used: MySQL, PHP and JavaScript

Regression Analysis using Portugal 2019 Election Results dataset

Sept. 2023 - Dec 2023

- The project is to predict how many MPs were elected at a district/national level after the 2019 Portugal Parliament Elections. To achieve proper results, we engineered Data Preprocessing techniques and regression models to predict district wise MP wins.
- Tools Used: Python (Pandas, Numpy, regression techniques), SQL

Skills

Languages: Python, SQL, Bash, PowerShell, R, C

Big Data & Cloud: AWS (EC2, S3, IAM), Snowflake, Google BigQuery, GCP, JFrog Artifactory

Data Engineering: ETL Development, Data Warehousing, Data Lakes, Airflow, MySQL Workbench

DevOps & CI/CD: Git, GitLab CI/CD, Jenkins, SonarQube, Shell Scripting

Project Management & Agile: Kanban boards, Scrum, CI/CD pipeline tracking

Tools & Frameworks: Pandas, NumPy, Scikit-Learn, TensorFlow, Jupyter, VS Code Confluence, Jira

Visualization: Tableau, Matplotlib, Seaborn

Operating Systems: Linux (Debian, Ubuntu), Windows

Certifications & Training

- Google Professional Data Engineer Certification. (in progress)
- MLOps with AWS Bootcamp - SageMaker, CI/CD, and cloud-native ML deployment (in progress)

Honors

- Best Productive Team Award in Recognition of the significant contribution to Teamcenter v12.x and PSI v16.x upgrade.